**Question 1a)**

The concentration of element ($M$) in 869 randomly selected pieces of glass is summarized below.

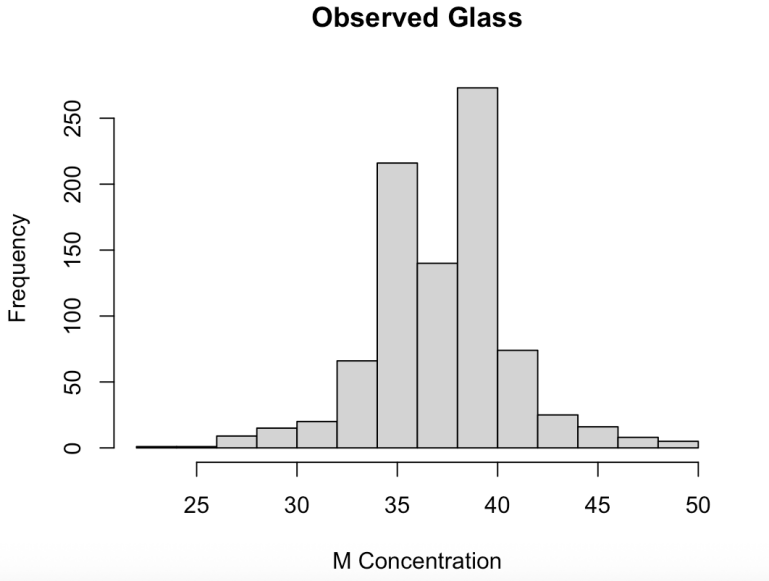



Figure 1: Frequency Histogram of Observed Element M Concentration

**Question 1b)**

A mixture prior is defined using the following weightings and parameters. A tau column is added for future reference.

| $i$ | $\gamma_i$ | $\mu_i$ | $\sigma_i^2$ | $\tau_i$ |
|---|---|---|---|---|
| 1 | 0.15 | 32.4 | 9 | 1/9 |
| 2 | 0.3 | 35.2 | 1 | 1 |
| 3 | 0.4 | 38.9 | 1 | 1 |
| 4 | 0.15 | 42.1 | 9 | 1/9 |

$$\pi(x) = \sum_{i=1}^{4} \gamma_i Normal(x; \mu_i, \sigma_i^2) \tag{1}$$

Superimposing the above prior function $\pi(x)$ on the normalized density histogram, we get the following graph.

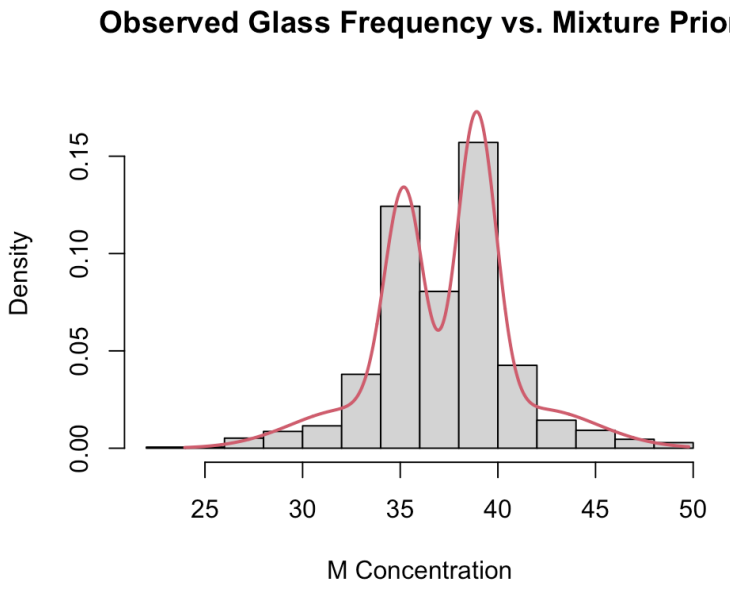



Figure 2: Frequency Histogram of Observed Element M Concentration

Visually comparing the mixture prior density function with the histogram in part a), it seems that the prior is a good fit to the observed data, as the bimodal behavior of the prior matches up with the observed data quite nicely.

**Question 1c)**

The likelihood function can be defined as $\pi(y|x) = Normal(y; x, 0.3^2)$ . Since we have not yet observed any data (i.e. measurements of $y$), we are here computing a prior predictive $\pi(y)$. Using a mixture prior will give a mixture prior predictive:

$$\pi(y) = \sum_{i=1}^{4} \gamma_i \pi_i(y) \tag{2}$$

Furthermore, if we have the likelihood function $\pi(y|x) = Normal(y; x, 0.3^2)$ and prior "pieces" $\pi_i(x) = Normal(x; \mu_i, \sigma_i^2)$, we can use normal-normal conjugacy to simplify the prior predictive for each of the four densities in the mixture as $\pi_i(y) = Normal(y; \mu_i, 0.3^2 + \sigma_i^2)$. Thus the prior predictive becomes

$$\pi(y) = \sum_{i=1}^{4} \gamma_i Normal(y; \mu_i, 0.3^2 + \sigma_i^2) \tag{3}$$

**Question 1d)**

To calculate the posterior $\pi(x|y)$, we can first find the posterior "pieces" $\pi_i(x|y)$, then update their weights.

$$\pi_i(x|y) = \frac{\pi(y|x)\pi_i(x)}{\pi_i(y)} \tag{4}$$

Since the likelihood $\pi(y|x) = Normal(y; x, 0.3^2)$ is normally distributed with known precision $\tau_0 = \frac{1}{0.3^2}$ and the prior "pieces" $\pi_i(x) = Normal(x; \mu_i, 1/\tau_i)$ are normally distributed, again we can use normal-normal conjugacy to compute the posterior as follows:

$$\pi_i(x|y) = Normal(x; \frac{\tau_0 y + \tau_i \mu_i}{\tau_0 + \tau_i}, \frac{1}{\tau_0 + \tau_i}) \tag{5}$$

Now, calculating the actual posterior by weighing each of the posterior "pieces":

$$\begin{aligned}
\pi(x|y) &= \sum_{i=1}^{4} \left( \frac{\gamma_i \pi_i(y)}{\sum_{j=1}^{4} \gamma_j \pi_j(y)} \right) \pi_i(x|y) \\
&= \sum_{i=1}^{4} \left( \frac{\gamma_i \pi_i(y)}{\sum_{j=1}^{4} \gamma_j \pi_j(y)} \right) Normal(x; \frac{\tau_0 y + \tau_i \mu_i}{\tau_0 + \tau_i}, \frac{1}{\tau_0 + \tau_i}) \\
&= \sum_{i=1}^{4} \gamma_i' Normal(x; \mu_i', \sigma_i'^2)
\end{aligned} \tag{6}$$

Where we define our new weights, means, and standard deviations as $\gamma_i'$, $\mu_i'$, and $\sigma_i'^2$ respectively. Computing for the new parameters for the posterior $\pi(x|y = 33.7)$, we have:

| $i$ | $\gamma_i'$ | $\mu_i'$ | $\sigma_i'^2$ |
|---|---|---|---|
| 1 | $3.048 \times 10^{-1}$ | 33.69 | 0.04972 |
| 2 | $6.883 \times 10^{-1}$ | 33.82 | 0.08257 |
| 3 | $1.057 \times 10^{-5}$ | 34.13 | 0.08257 |
| 4 | $6.900 \times 10^{-3}$ | 33.78 | 0.04972 |

Looking at the weightings, it seems that the first and second second "pieces" of the mixture prior is now weighted more, and the third and fourth "pieces" of the mixture prior are now weighted much less. This makes sense due to the fact that the observation of $y = 33.7$ is much closer to the $\pi_1(x)$ and $\pi_2(x)$ prior means of 32.4 and 35.2, while the $\pi_3(x)$ and $\pi_4(x)$ prior means of 38.9 and 42.1 are much further away. The posterior $\pi(x|y)$ now looks much more like a single normal distribution, and is graphed below. The observation of $y = 33.7$ indicated on the graph.

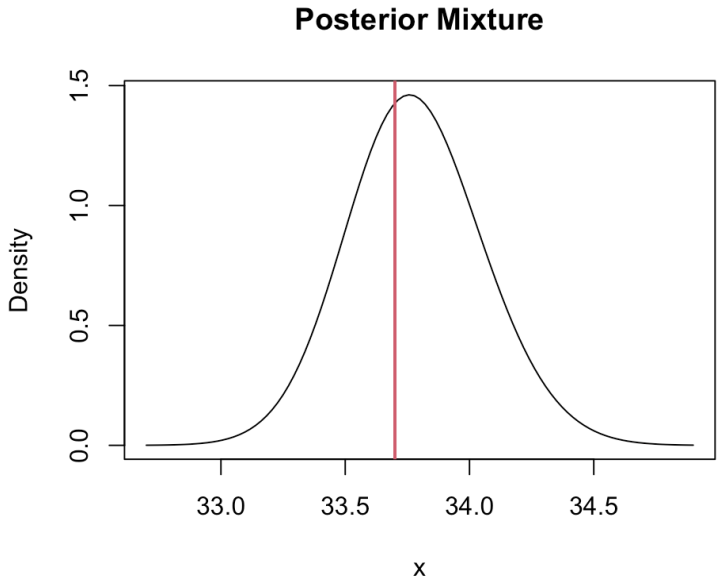



Figure 3: Posterior Mixture

**Question 1e)**

Now we are computing a posterior predictive for $y_{NEW}$, with our "posterior-prior" being the mixture distribution after one observation of $y = 33.7$. As the likelihood is still distributed with density function $\pi(y_{NEW}|x) = Normal(y; x, 0.3^2)$, we again can use the same normal-normal conjugacy technique as used in part 1c) to compute the posterior predictive as follows:

$$\begin{aligned} \pi(y_{NEW}|y) &= \sum_{i=1}^{4} \gamma_i' \pi_i(y_{NEW}|y) \\ &= \sum_{i=1}^{4} \gamma_i' Normal(y_{NEW}; \mu_i', 0.3^2 + \sigma_i'^2) \end{aligned} \tag{7}$$

The graph of the posterior predictive is below. It is very similar to the posterior mixture graph, with the difference being the wider shape. This is to be expected, due to the greater variance added from the variability in observing the true value $x$ from the likelihood function.

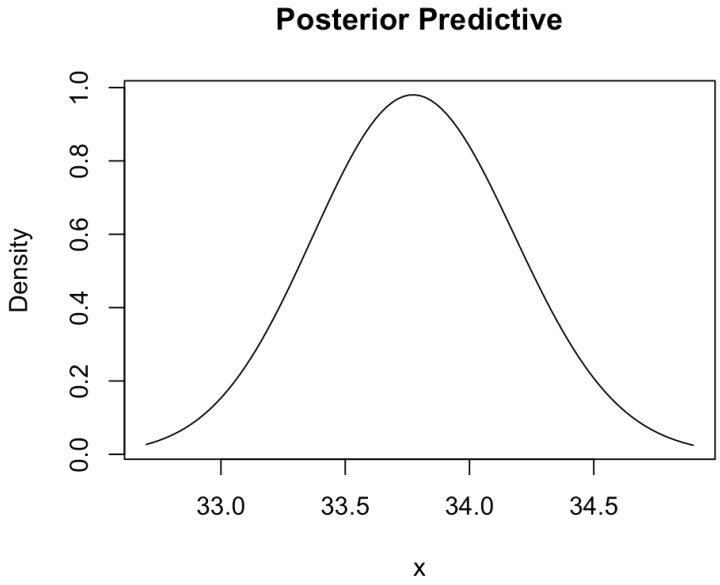



Figure 4: Posterior Predictive

**Question 1f)**

After making measurements $y_s$ from the suspect and $y_c$ from the crime scene, we can calculate the likelihood ratio defined as:

$$\begin{aligned} LR &= \frac{\pi(data|H_p)}{\pi(data|H_d)} \\ &= \frac{\pi(y_s, y_c|y_s = y_c)}{\pi(y_s, y_c|y_s \neq y_c)} \end{aligned} \tag{8}$$

Given $H_p$ where the glass measurements from both $y_s$ and $y_c$ come from the same source with actual concentration $x$, the probability density of the numerator is thus the prior predictive of observing $y_s$ multiplied by the posterior predictive of observing $y_c$ given $y_s$ as defined in equation (7).

Given $H_d$ where the glass measurements from $y_s$ and $y_c$ come from independent sources, the two measurements will come from sources with two different underlying concentrations $x$. Therefore in the denominator, both $y_s$ and $y_c$ will be distributed independently following the prior predictive distribution as defined in equation (3)

The likelihood ratio $LR$ can thus be written as:

$$\begin{aligned} LR &= \frac{\pi(y_s)\pi(y_c|y_s)}{\pi(y_c)\pi(y_s)} \\ &= \frac{\pi(y_c|y_s)}{\pi(y_c)} \\ &= \frac{\sum_{i=1}^{4} \gamma_i' Normal(y_c; \mu_i', 0.3^2 + \sigma_i'^2)}{\sum_{i=1}^{4} \gamma_i Normal(y_c; \mu_i, 0.3^2 + \sigma_i^2)} \end{aligned} \tag{9}$$

**Question 1g)**

The observed glass concentration on the suspect and at the crime scene are $y_s = 33.7$ and $y_c = 33.9$. Computing the posterior predictive and two prior predictives we get:

$$\begin{aligned} \pi(y_c = 33.9|y_s = 33.7) &= 0.9336 \\ \pi(y_c = 33.9) &= 0.07083 \end{aligned} \tag{10}$$

The likelihood ratio $LR$ therefore becomes:

$$\begin{aligned} LR &= \frac{\pi(y_c|y_s)}{\pi(y_s)} \\ &= \frac{0.9336}{0.07083} \\ &= 13.18 \end{aligned} \tag{11}$$

Without any presumptions on the probabilities of whether the glass from the suspect and from the crime scene are from the same or different sources[1], we can set the odds ratio as $\frac{\pi(H_p)}{\pi(H_d)} = 1$. Thus the posterior odds ratio can be calculated to be:

$$\begin{aligned} \frac{\pi(H_p|data)}{\pi(H_d|data)} &= \frac{\pi(data|H_p)}{\pi(data|H_d)} \times \frac{\pi(H_p)}{\pi(H_d)} \\ &= 13.18 \times 1 \\ &= 13.18 \end{aligned} \tag{12}$$

The following conclusion can then be drawn:

Without presumptions on the probabilities of whether the glass from the suspect and from the crime scene are from the same or different sources, given the data and proposed methodology, it is more than 13 times more likely that the glass from the suspect and the crime scene are from the same source compared with the alternative where the glass from the suspect and crime scene are from different sources.

[1]Perhaps in a society whose legal process is defined on a presumption of innocence, the suspect will have a more lenient odds ratio...

**Question 2a)**

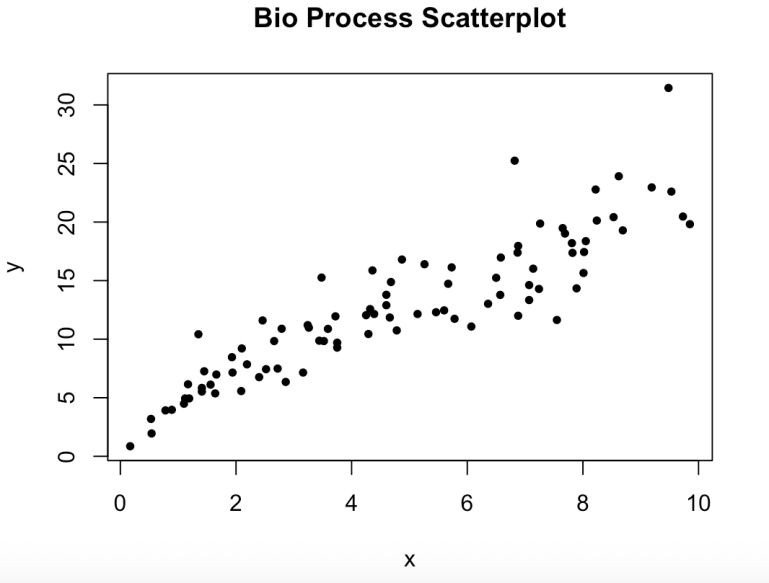



Figure 5: Scatterplot of Bio Process

**Question 2b)**

The likelihood function for each Bio Process $y_i$ given its associated $x_i$ and parameters $\alpha, \beta, \gamma$ is defined as follows:

$$\pi(y_i|x_i, \alpha, \beta, \gamma) = Gamma(y_i; \alpha, \frac{\alpha}{\beta x_i + \gamma \log(x_i + 1)}) \tag{13}$$

Fixing $\alpha = 25$, the likelihood function for individual observations $(x_i, y_i)$ becomes:

$$\pi(y_i|x_i, \beta, \gamma) = Gamma(y_i; 25, \frac{25}{\beta x_i + \gamma \log(x_i + 1)}) \tag{14}$$

Assuming all 89 observations $(x_i, y_i)$ are independently distributed, the likelihood function is therefore:

$$\pi(y|x, \beta, \gamma) = \prod_{i=1}^{89} Gamma(y_i; 25, \frac{25}{\beta x_i + \gamma \log(x_i + 1)}) \tag{15}$$

Since the likelihood is an extremely small product of 89 small numbers and thus difficult to work with, the log-likelihood is computed as below:

$$\begin{aligned}\log \pi(y|x,\beta,\gamma) &= \log\left[\prod_{i=1}^{89} Gamma(y_i; 25, \frac{25}{\beta x_i + \gamma \log(x_i+1)}\right] \\ &= \sum_{i=1}^{89} \log\left[Gamma(y_i; 25, \frac{25}{\beta x_i + \gamma \log(x_i+1)}\right]\end{aligned} \tag{16}$$

Implementing the code for equation (16) in R to find the logarithm of the likelihood of the data given values for the parameters $\beta$ and $\gamma$ as input, we have:

```r
# Computes log-likelihood using bio_x data given input params beta and gamma
log_bio_likelihood <- function(beta, gamma){
  sum(log(dgamma(bio_y, 25, 25/(beta * bio_x + gamma * log(bio_x + 1)))))
}

# Vectorizes the likelihood functions so vectors of beta/gamma can be inputted
log_bio_likelihood <- Vectorize(log_bio_likelihood)
```

Figure 6: Code Snippet for Computing Log-Likelihood in R

**Question 2c), 2d)**

The posterior function for individual observations $x_i, y_i$ is proportional to the likelihood function times the prior function. Also, due to the independence of parameters in the prior and the fact that both $\alpha$ and $x_i$ are given for each observed $y_i$, we can further simplify the posterior as follows:

$$\begin{aligned}\pi(x_i,\alpha,\beta,\gamma|y_i) &\propto_{x_i,\alpha,\beta,\gamma} \pi(y_i|x_i,\alpha,\beta,\gamma)\cdot\pi(x_i,\alpha,\beta,\gamma) \\ &\propto_{x_i,\alpha,\beta,\gamma} \pi(y_i|x_i,\alpha,\beta,\gamma)\cdot\pi(x_i)\cdot\pi(\alpha)\cdot\pi(\beta)\cdot\pi(\gamma) \\ &\propto_{\beta,\gamma} \pi(y_i|\beta,\gamma)\cdot\pi(\beta)\cdot\pi(\gamma)\end{aligned} \tag{17}$$

The posterior can thus be calculated using the previous results from equation (16):

$$\begin{aligned}\pi(\beta,\gamma|y) &\propto_{\beta,\gamma} \pi(y|\beta,\gamma)\pi(\beta)\pi(\gamma) \\ &\propto_{\beta,\gamma} \exp\left[\sum_{i=1}^{89} \log\left[Gamma(y_i; 25, \frac{25}{\beta x_i + \gamma \log(x_i+1)}\right]\right]\frac{1}{\beta\gamma}\end{aligned} \tag{18}$$

The code below was used to implement the posterior calculations. A discretized grid given lower and upper bounds for $\beta$ and $\gamma$ was first defined. Then, for each $\beta$ and $\gamma$ pair in the grid, the posterior was calculated by applying equation (18). Finally, a heat map for the posterior was produced by looking at for which $\beta$ and $\gamma$ pairs on the grid, the posterior density was high.

```r
# Calculates posterior given beta and gamma values
bio_posterior <- function(beta, gamma){
  # exp(log_bio_likelihood(beta, gamma)) / (beta * gamma)
  exp(log_bio_likelihood(beta, gamma) - log(beta) - log(gamma))
}

# Vectorizes the posterior
bio_posterior <- Vectorize(bio_posterior)

# Setting up bounds for beta and gamma values and number of discretizations
beta_0 <- 0.3
beta_1 <- 1.3
gamma_0 <- 4
gamma_1 <- 7
discretization <- 100

# Computes vectors of beta, gamma using the given bounds
beta_range <- seq(beta_0, beta_1, length.out = discretization)
gamma_range <- seq(gamma_0, gamma_1, length.out = discretization)

# Outer product of beta gamma pairs, and applies bio_posterior to get a matrix
# of posterior values corresponding to the coordinates on the grid
bio_posterior_grid <- outer(beta_range, gamma_range, bio_posterior)

# Improves numerics? From lecture 3 code
bio_posterior_grid <- bio_posterior_grid - max(bio_posterior_grid)

# Re-scaling the posterior to sum to 1
bio_posterior_grid <- bio_posterior_grid/sum(bio_posterior_grid)

# Heat map
image(beta_range, gamma_range, bio_posterior_grid, main="Posterior Distribution", xlab="Beta", ylab="Gamma")
```

Figure 7: Code Snippet for Computing Posterior

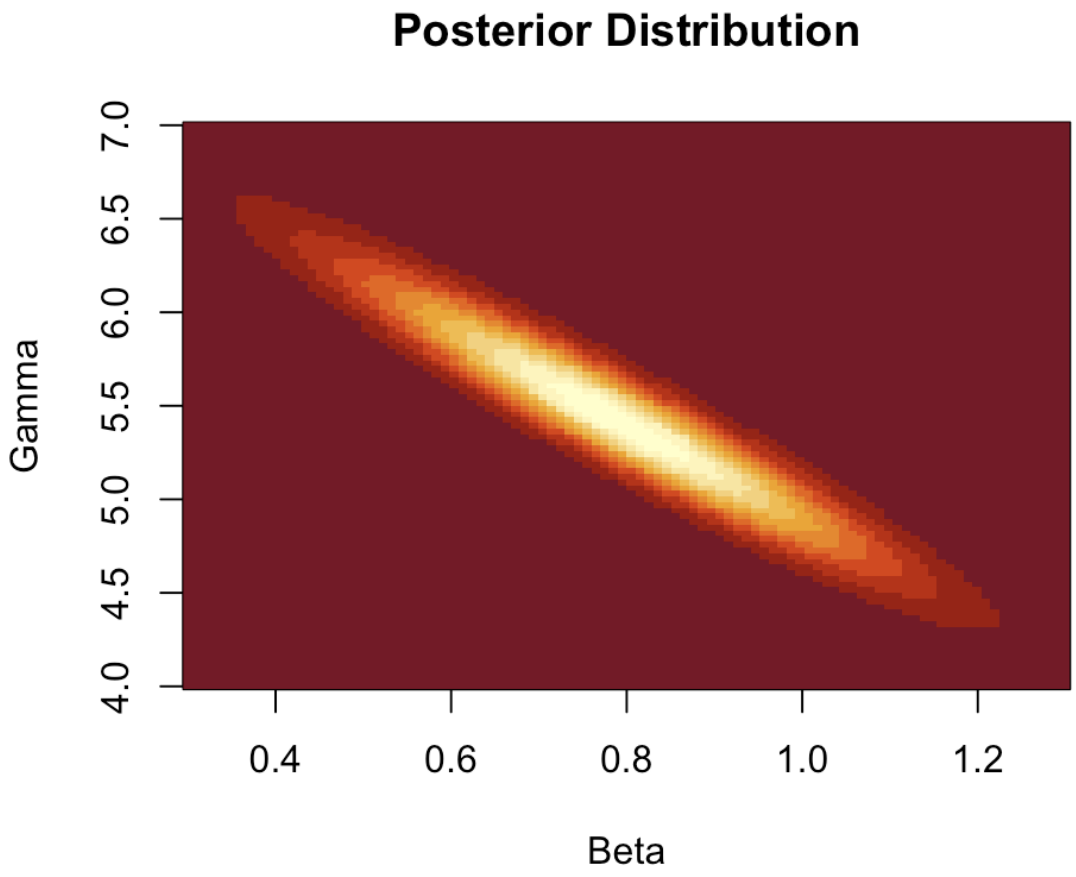



Figure 8: Posterior Heat Map

From the heat map in Figure 8 above, roughly only in the range of $\beta \in [0.3, 1.3]$ and $\gamma \in [4, 7]$ that we have values of the posterior that are non-negligible.

**Question 2e)**

To compute the approximate probability that an output $y$ will be greater than 10 when a new input is $x = 10$, we can first find the posterior predictive:

$$\begin{aligned} \pi(y|y_{DATA}, x = 10) &= \iint \pi(y|\beta, \gamma, x = 10) \cdot \pi(\beta, \gamma|y_{DATA}) d\beta d\gamma \\ &\approx \sum_{\beta} \sum_{\gamma} \pi(y|\beta, \gamma, x = 10) \cdot \pi(\beta, \gamma|y_{DATA}) \\ &\approx \sum_{\beta} \sum_{\gamma} Gamma(y; 25, \frac{25}{10\beta + \gamma \log(11)}) \cdot \pi(\beta, \gamma|y_{DATA}) \end{aligned} \tag{19}$$

Then, we can find the probability that $y > 10$ by integrating over the likelihood Gamma density:

$$\begin{aligned} Pr(y > 20|y_{DATA}, x = 10) &\approx \sum_{\beta}\sum_{\gamma}\int_{20}^{\infty} Gamma(y; 25, \frac{25}{10\beta + \gamma\log(11)})dy \cdot \pi(\beta, \gamma|y_{DATA}) \\ &\approx 0.5459 \end{aligned} \tag{20}$$

**Question 2f)**

In a more general model where $\alpha$ is a parameter with prior $\pi(\alpha) \propto \frac{1}{\alpha}$, we can again first utilize a discretization over a 3D grid to approximate over selected ranges of the three variables $\alpha, \beta, \gamma$. After computing the approximated log-posterior density, instead of visually selecting the appropriate region over a 3D grid to integrate over for calculating the probability, we can use the Laplace approximation to find a multivariate normal distribution that approximates the log-posterior density. The MAP estimates for $\alpha, \beta, \gamma$ using the Laplace approximation, as well as the covariance matrix is presented below.

$$\hat{\beta} = 0.7944, \hat{\gamma} = 5.409, \hat{\alpha} = 27.38 \tag{21}$$

$$K_{\beta,\gamma,\alpha} = \begin{bmatrix} -0.03403 & -0.08620 & 0.01661 \\ -0.08620 & 0.2411 & -0.03940 \\ 0.01661 & -0.03940 & 17.03 \end{bmatrix} \tag{22}$$

Now using these MAP estimates and the assumption that the multivariate normal is a good approximation of the original densities near the MAP, we can further make predictions about the data.